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Display module and display device

Abstract

A display module and a display device are provided. The display module includes a display panel, a flexible circuit board, and a printed circuit board. The display panel includes a bonding area. The flexible circuit board and the printed circuit board are electrically bonded and connected to the display panel in the bonding area, and are connected by a connector. The connector includes a first connection board and a second connection board. A side of the first connection board includes a groove including first pins. A side of the second connection board facing the first connection board includes second pins. The first pins and the second pins have a one-to-one correspondence. The second pins are plugged into the grooves, and are in contact with and electrically connected to the first pins; and a portion of at least one of the first pins is coated with a first insulating layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) This application claims the priority of Chinese Patent Application No. 202211192012.X, filed on Sep. 28, 2022, the content of which is incorporated herein by reference in its entirety.
TECHNICAL FIELD
(2) The present disclosure generally relates to the field of display technologies and, more particularly, relates to a display module and a display device.
BACKGROUND
(3) With rapid development of flat panel displays and improvement of people's living standards, display devices are more and more widely used in people's daily life. A connector is an indispensable part of a display device. The connector is mainly used to set up a bridge of communication between blocked or isolated circuits in circuits of the display device, such that current can flow and the circuit can achieve predetermined functions. Forms and structures of connectors are widely different, and there are various forms of connectors with different application objects, frequencies, power and application environments.
(4) In existing display devices, connectors used to provide the signal communication between a circuit board for providing driving signals and a display panel are mainly connectors with plug-in structures. Electrical connection between the display panel and a power supply can be conveniently

and quickly completed by using the form of plug-in connectors, which is beneficial to industrialized mass production. However, in the current conventional design of connectors, and metallic conductive pins are exposed, and there is no dust-proof effect. It is prone to the problem of misplaced plugging and unplugging during plugging and unplugging with electricity. Once the improper operation of misplaced plugging and plugging occurs, it is easy to cause static electricity and damage the panel. The panel quality and product yield may be affected. Also, the operator's brute force dislocation and oblique insertion may easily cause problems such as connector damage.
(5) Therefore, to provide a display module and a display device that can avoid electrostatic damage to the panel during the process of plugging and unplugging the connector, to protect equipment and improve product yield, is a technical problem to be solved.
SUMMARY
(6) One aspect of the present disclosure provides a display module. The display module includes: a display panel, a flexible circuit board, and a printed circuit board. The display panel includes a bonding area. The flexible circuit board and the printed circuit board are electrically bonded and connected to the display panel in the bonding area. The flexible circuit board and the printed circuit board are connected by a connector. The connector includes a first connection board and a second connection board. A side of the first connection board facing the second connection board includes a groove. The groove includes a plurality of first pins. A side of the second connection board facing the first connection board includes a plurality of second pins. The plurality of first pins and the plurality of second pins have a one-to-one correspondence. The plurality of second pins is plugged into the grooves, and is in contact with and electrically connected to the plurality of first pins; and a portion of at least one of the plurality of first pin is coated with a first insulating layer.
(7) Another aspect of the present disclosure provides a display device. The display device includes a display module. The display module includes a display panel, a flexible circuit board, and a printed circuit board. The display panel includes a bonding area. The flexible circuit board and the printed circuit board are electrically bonded and connected to the display panel in the bonding area. The flexible circuit board and the printed circuit board are connected by a connector. The connector includes a first connection board and a second connection board. A side of the first connection board facing the second connection board includes a groove. The groove includes a plurality of first pins. A side of the second connection board facing the first connection board includes a plurality of second pins. The plurality of first pins and the plurality of second pins have a one-to-one correspondence. The plurality of second pins is plugged into the grooves, and is in contact with and electrically connected to the plurality of first pins; and a portion of at least one of the plurality of first pin is coated with a first insulating layer.
(8) Other aspects or embodiments of the present disclosure can be understood by those skilled in the art in light of the description, the claims, and the drawings of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The following drawings are merely examples for illustrative purposes according to various disclosed embodiments and are not intended to limit the scope of the present disclosure.
(2) FIG. **1** illustrates a planar structure of an exemplary display module consistent with various disclosed embodiments in the present disclosure;
(3) FIG. **2** illustrates a connecting structure of a flexible circuit board and a printed circuit board in FIG. **1** consistent with various disclosed embodiments in the present disclosure;
(4) FIG. **3** illustrates an exploded structure of a cross-sectional view of a connector in FIG. **2**, consistent with various disclosed embodiments in the present disclosure;
(5) FIG. **4** illustrates a three-dimensional structure of a first connection board in FIG. **3**, consistent

with various disclosed embodiments in the present disclosure;
(6) FIG. **5** illustrates a locally enlarged view of a Q region in FIG. **4** consistent with various disclosed embodiments in the present disclosure;
(7) FIG. **6** illustrates a top-view structure of a first connection board in FIG. **4** consistent with various disclosed embodiments in the present disclosure;
(8) FIG. **7** illustrates a cross-sectional view along an A-A′ direction in FIG. **6** consistent with various disclosed embodiments in the present disclosure;
(9) FIG. **8** illustrates another cross-sectional view along an A-A′ direction in FIG. **6** consistent with various disclosed embodiments in the present disclosure;
(10) FIG. **9** illustrates an exploded view of cross-section structures of a portion of first sub-pins and second sub-pins in a connector in FIG. **2** consistent with various disclosed embodiments in the present disclosure;
(11) FIG. **10** illustrates an exploded view of cross-section structures of a third sub-pin and a fourth sub-pin in a connection in FIG. **2** consistent with various disclosed embodiments in the present disclosure;
(12) FIG. **11** illustrates a schematic structure when the first connection board and the second connection board in FIG. **9** and FIG. **10** are misaligned and plugged consistent with various disclosed embodiments in the present disclosure;
(13) FIG. **12** illustrates another schematic structure when the first connection board and the second connection board in FIG. **9** and FIG. **10** are misaligned and plugged consistent with various disclosed embodiments in the present disclosure;
(14) FIG. **13** illustrates another exploded view of cross-section structures of a portion of first sub-pins and second sub-pins in a connector in FIG. **2** consistent with various disclosed embodiments in the present disclosure;
(15) FIG. **14** illustrates another exploded view of cross-section structures of a portion of first sub-pins and second sub-pins in a connector in FIG. **2** consistent with various disclosed embodiments in the present disclosure;
(16) FIG. **15** illustrates another exploded view of cross-section structures of a portion of first sub-pins and second sub-pins in a connector in FIG. **2** consistent with various disclosed embodiments in the present disclosure;
(17) FIG. **16** illustrates another exploded view of cross-section structures of a portion of third sub-pins and fourth sub-pins in a connector in FIG. **2** consistent with various disclosed embodiments in the present disclosure;
(18) FIG. **17** illustrates a schematic structure when the first connection board and the second connection board in FIG. **9** and FIG. **16** are misaligned and plugged consistent with various disclosed embodiments in the present disclosure;
(19) FIG. **18** illustrates another schematic structure when the first connection board and the second connection board in FIG. **9** and FIG. **16** are misaligned and plugged consistent with various disclosed embodiments in the present disclosure;
(20) FIG. **19** illustrates another exploded view of cross-section structures of a portion of first sub-pins and second sub-pins in a connector in FIG. **2** consistent with various disclosed embodiments in the present disclosure;
(21) FIG. **20** illustrates another exploded view of cross-section structures of a portion of third sub-pins and fourth sub-pins in a connector in FIG. **2** consistent with various disclosed embodiments in the present disclosure;
(22) FIG. **21** illustrates another exploded view of cross-section structures of a portion of third sub-pins and fourth sub-pins in a connector in FIG. **2** consistent with various disclosed embodiments in the present disclosure;
(23) FIG. **22** illustrates another exploded view of cross-section structures of a portion of third sub-pins and fourth sub-pins in a connector in FIG. **2** consistent with various disclosed embodiments in

the present disclosure;
(24) FIG. **23** illustrates another planar structure of a display module consistent with various disclosed embodiments in the present disclosure;
(25) FIG. **24** illustrates a connecting structure of a flexible circuit board and a printed circuit board in FIG. **23** consistent with various disclosed embodiments in the present disclosure;
(26) FIG. **25** illustrates another connecting structure of a flexible circuit board and a printed circuit board in FIG. **23** consistent with various disclosed embodiments in the present disclosure;
(27) FIG. **26** illustrates a three-dimensional exploded structure of a connector in FIG. **23** consistent with various disclosed embodiments in the present disclosure;
(28) FIG. **27** illustrates an exemplary display device consistent with various disclosed embodiments in the present disclosure.

DETAILED DESCRIPTION

(29) Reference will now be made in detail to exemplary embodiments of the disclosure, which are illustrated in the accompanying drawings. Hereinafter, embodiments consistent with the disclosure will be described with reference to drawings. In the drawings, the shape and size may be exaggerated, distorted, or simplified for clarity. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts, and a detailed description thereof may be omitted.

(30) Further, in the present disclosure, the disclosed embodiments and the features of the disclosed embodiments may be combined under conditions without conflicts. It is apparent that the described embodiments are some but not all of the embodiments of the present disclosure. Based on the disclosed embodiments, persons of ordinary skill in the art may derive other embodiments consistent with the present disclosure, all of which are within the scope of the present disclosure.

(31) Moreover, the present disclosure is described with reference to schematic diagrams. For the convenience of descriptions of the embodiments, the cross-sectional views illustrating the device structures may not follow the common proportion and may be partially exaggerated. Besides, those schematic diagrams are merely examples, and not intended to limit the scope of the disclosure. Furthermore, a three-dimensional (3D) size including length, width, and depth should be considered during practical fabrication.

(32) The present disclosure provides a display module. FIG. **1** is a schematic planar view of a display module provided by one embodiment of the present disclosure, FIG. **2** is a schematic diagram of a connection structure of a flexible circuit board and a printed circuit board in FIG. **1**, FIG. **3** is an exploded structure of the cross-sectional structure of a connector in FIG. **2**, FIG. **4** is a schematic three-dimensional structural diagram of a first connection board in FIG. **3**, and FIG. **5** is a partially enlarged structural schematic diagram of the Q area in FIG. **4**. As shown in FIG. **1** to FIG. **5**, in one embodiment, the display module **000** may include a display panel **10**, a flexible circuit board **20**, and a printed circuit board **30**. The display panel **10** may include a bonding area BA, and the flexible circuit board **20** and the printed circuit board **30** may be bonded and electrically connected to the display panel **10** in the bonding area BA.

(33) The flexible circuit board **20** and the printed circuit board **30** may be connected through a connector **40**.

(34) The connector **40** may include a first connection board **401** and a second connection board **402**.

(35) A side of the first connection board **401** facing the second connection board **402** may be provided with a groove **4010**, and a plurality of first pins **4011** may be disposed in the groove **4010**.

(36) A side of the second connection board **402** facing the first connection board **401** may be provided with a plurality of second pins **4021**. The plurality of first pins **4011** and the plurality of second pins **4021** may have one-to-one correspondence. The plurality of second pins **4021** may be plugged into the grooves **4010** and connected. The plurality of second pin **4021** may be in contact and electrically connected with the plurality of first pins **4011**.

(37) A partial area of at least one of the plurality of first pins **4011** may be coated with a first insulating layer **50**.

(38) Specifically, in the present embodiment, the display module **000** may include the display panel **10**, the flexible circuit board **20**, and the printed circuit board **30**. The flexible circuit board **20** and the printed circuit board **30** may be bonded and electrically connected to the display panel **10** in the bonding area BA of the display panel **10**. Optionally, the flexible circuit board **20** may be bonded to the display panel **10** in the bonding area BA of the display panel **10**. The flexible circuit board may be connected to **20** the printed circuit board **30** through the connector **40**. Correspondingly, the printed circuit board **30** may provide the display panel **10** with electrical signals that drive the display panel **10** to realize functions such as display through the lines on the flexible circuit board **20**.

(39) In one embodiment, the printed circuit board **30** may be a provider of electrical connections for electronic components. Optionally, the printed circuit board **30** may include a substrate, and electronic components or a drive circuit integrated on the substrate. The flexible circuit board **20** (FPC) may be a highly reliable and excellent flexible printed circuit made of polyimide or polyester film as the base material plate. The display module **000** of this embodiment may include the display panel **10** and the printed circuit board **30**. The electrical connection between the display panel **10** and the printed circuit board **30** may be generally connected by the flexible circuit board **20**. The flexible circuit board **20** may be bonded to the display panel **10** to realize electrical signal transmission with the display panel **10**. The electrical signal transmission between the flexible circuit board **20** and the printed circuit board **30** may be realized by plugging and connecting to the connector **40**. The flexible circuit board **20** may generally include a pressing end **20**A and a gold finger end **20**B. The pressing end **20**A of the flexible circuit board **20** may be pressed within the range of the bonding area BA of the display panel **10** to realize electrical connection with the display panel **10**. At the same time, the connector **40** connected to the printed circuit board **30** may be disposed at the gold finger end **20**B of the flexible circuit board **20**. The connector **40** of this embodiment may include the first connection board **401** and the second connection board **402**. Optionally, the first connection board **401** may be disposed at the position of the gold finger end **20**B of the flexible circuit board **20** or at a corresponding position on the printed circuit board **30**. The second connection board **402** may be disposed at the position of the gold finger end **20**B of the flexible circuit board **20** or at the another one of corresponding positions on the printed circuit board **30**. The side of the first connection board **401** facing the second connection board **402** may be provided with the groove **4010**. The plurality of first pins **4011** may be disposed in the groove **4010**. The side of the second connection board **402** facing the first connection board **401** may be provided with the plurality of second pins **4021**. The plurality of first pins **4011** and the plurality of second pins **4021** may have one-to-one correspondence.

(40) Optionally, the depth of the groove **4010** may not penetrate through the thickness of the first connection board **401**, and the plurality of first pins **4011** may be disposed on the inner sidewall of the groove **4010** for subsequent plugging between the first connection board **401** and the second connection board **402**. The plurality of second pins **4021** on the second connection board **402** may be plugged into the grooves **4010** to contact and electrically connect with the plurality of first pins **4011**. Therefore, the electrical connection between the first connection board **401** and the second connection board **402** may be realized through the contact between the plurality of first pins **4011** on the first connection board **401** and the plurality of second pins **4021** on the second connection board **402**, enabling the transmission of electrical signals between the driving circuit board **30** and the flexible circuit board **20**. In this embodiment, the electrical connection between the flexible circuit board **20** and the printed circuit board **30** may be realized through the plugging and unplugging between the first connection board **401** and the second connection board **402** of the connector **40**. Therefore, the electrical connection between the display panel **10** and the printed circuit board **30** may be realized through the flexible circuit board **20**. In the present disclosure, a

structure in which the flexible circuit board **20** and the printed circuit board **30** are electrically connected through the plugging and unplugging of the connector **40**. The electrical connection between the display panel **10** and the printed circuit board **30** may be able to easily and quickly completed, which is beneficial to the industrialized mass production of the module.

(41) However, in the existing technologies, pins on the connector generally use a conventional conductive design. The metal of the pins is exposed, and there is no fool-proofing effect. When plugging and unplugging with electricity occur, it is easy to cause static electricity damage to the display panel during the operation of plugging and unplugging with electricity. Operations of brute force with misalignment and oblique plugging could easily cause damage to the connector, which may induce poor yield of the module.

(42) In the present embodiment, a partial area of at least one first pin **4011** of the plurality of first pins **4011** in the first connection board **401** may be coated with the first insulating layer **50**. Optionally, in one embodiment, a partial area of each of the plurality of first pins **4011** in the first connection board **401** may be coated with the first insulating layer **50**. Therefore, when the flexible circuit board **20** and the printed circuit board **30** are electrically connected through the connector **40**, the plurality of second pins **4021** on the second connection board **402** may be plugged into the grooves **4010** opened in the first connection board **401**, and may be electrically connected to the corresponding plurality of first pins **4011**. Since a partial area of each of the plurality of first pins **4011** in the first connection board **401** may be coated with the first insulating layer **50**, electrostatic breakdown because of the instantaneous contact between the plurality of first pins **4011** and the plurality of second pins **4021** during the instantaneous wrong plugging may be prevented. Therefore, electrostatic damage to the display panel **10** induced by plugging and unplugging of the connector **40** may be avoided. Further, since the first insulating layer **50** in this embodiment is not coated on the entire surface of the plurality of first pins **4011**, the conductivity of the plurality of first pins **4011** may not be affected. Therefore, when the first connection board **401** and the second connection board **402** are plugged, the plurality of second pins **4021** and the areas of the plurality of first pins **4011** not coated with the first insulating layer **50** may still be guaranteed to contact to realize conductive transmission between the first connection board **401** and the second connection board **402**. In this embodiment, the first insulating layer **50** may have been coated on a partial area of at least one of the plurality of first pins **4011** during the manufacturing process of the connector **40**. There may be no need to remove the first insulating layer **50** after the first connection board **401** and the second connection board **402** are plugged and connected, which may be beneficial to reduce the overall process steps of the module.

(43) The present disclosure does not specifically limit the type and structure of the display panel **10**. During specific implementation, the solution of this embodiment may be applied to any type of display panel **10**, as long as the connector **40** uses the design structure of this embodiment when the driving signal on the display panel **10** needs to realize electrical transmission through the flexible circuit board **20** and the driving circuit board **30**. The structure of the display panel **10** is not described in detail in this embodiment.

(44) For description purposes only, the drawings in this embodiment are merely illustrative of the shapes and numbers of the plurality of first pins **4011** and the plurality of second pins **4021** in the connector **40**, and are used as examples only to illustrate the present disclosure. In various embodiments, during specific implementation, the shapes and numbers of the plurality of first pins **4011** and the plurality of second pins **4021** in the connector **40** are not limited to this, and other setting methods are also possible, as long as only a partial area of the surfaces of the plurality of first pins **4011** in the first connection board **401** is coated with the first insulating layer **50** to avoid the occurrence of electrostatic damage because of operation with electricity of the connector **40** in the process of aligning plugging and unplugging, which are not limited in this embodiment.

(45) In some embodiments as shown in FIG. **1** to FIG. **5**, FIG. **6** which is a top view of the first connection board in FIG. **4**, and FIG. **7** which is a cross-sectional view along the A-A′ direction in

FIG. **6**, a shape of an orthographic projection of the groove **4010** of the first connection board **401** on a plane where the first connection board **401** is located may include a long strip.

(46) The groove **4010** may include a first side wall **4010**A and a second side wall **4010**B. One first pin **4011** of the plurality of first pins **4011** may include a first sub-section **4011**A and a second sub-section **4011**B, where the first sub-section **4011**A may be arranged on the first side wall **4010**A and the second sub-section **4011**B may be arranged on the second side wall **4010**B.

(47) In a direction Z perpendicular to the plane of the first connection board **401**, the first sub-section **4011**A and/or the second sub-section **4011**B may include a first region **4011**X and a second region **4011**Y, where the first region **4011**X is located on a side of the second region **4011**Y close to the bottom of the groove **4010**.

(48) Along a direction X from the first sub-section **4011**A to the second sub-section **4011**B, the thickness D**1** of the first sub-section **4011**A in the first region **4011**X may be smaller than a thickness D**2** of the first sub-section **4011**A in the second region **4011**Y; and/or, along the direction X from the first sub-section **4011**A to the second sub-section **4011**B, a thickness D**3** of the second sub-section **4011**B in the first region **4011**X may be smaller than a thickness D**4** of the second sub-section **4011**B in the second region **4011**Y.

(49) In the present embodiment, the groove **4010** may be provided at the side of the first connection board **401** facing the second connection board **402**, and the groove **4010** may be a groove structure formed by recession of a surface of the first connection board **401** facing the second connection board **402** in a direction away from the second connection board **402**. In the present embodiment, the shape of the orthographic projection of the groove **4010** on the plane where the first connection board **401** is located may be a long strip. In some other embodiment, the shape of the orthographic projection of the groove **4010** on the plane where the first connection board **401** is located may also be a closed ring or square, which is not limited in this embodiment. The present embodiment shown in drawings where the orthographic projection of the groove **4010** on the plane where the first connection board **401** is located is a rectangle is used as an example for illustration. The groove **4010** of the first connection board **401** may include the first side wall **4010**A and the second side wall **4010**B. Optionally, as shown in FIG. **3**, the groove **4010** may be understood as including a groove bottom **4010**C and a groove bottom **4010**C, and the first sidewall **4010**A and the second sidewall **4010**B at two opposite sides of the groove bottom **4010**C respectively. The plane where the first sidewall **4010**A is located may be understood as intersecting or perpendicular to the plane where the first connection board **401** is located, and the plane where the second sidewall **4010**B is located may be understood that it intersects or is perpendicular to the plane where the first connection board **401** is located. The plurality of first pins **4011** of the first connection board **401** may be disposed on two sidewalls of the groove **4010**. One first pin **4011** of the plurality of first pins **4011** of the first connection board **401** may include a first sub-section **4011**A and a second sub-section **4011**B. The first sub-section **4011**A and the second sub-section **4011**B may be understood as independent pieces disposed on the two sidewalls of the groove **4010**. The first sub-section **4011**A may be disposed on the first side wall **4010**A, and the second sub-section **4011**B may be disposed on the second side wall **4010**B. The first sub-section **4011**A and the second sub-section **4011**B may together form the one first pin **4011** of the first connection board **401**. When the second connection board **402** is plugged into the first connection board **401**, each of the plurality of second pins **4021** of the second connection board **402** may be in contact with the first sub-section **4011**A and the second sub-section **4011**B of a corresponding one of the plurality of first pins **4011** in the groove **4010**, to realize electrical connection between the corresponding pins of the first connection board **401** and the second connection board **402**.

(50) In the present embodiment, along the direction Z perpendicular to the plane where the first connection board **401** is located, the first sub-section **4011**A and/or the second sub-section **4011**B may include the first region **4011**X and the second region **4011**Y with different thicknesses. That is, in the direction Z perpendicular to the plane where the first connection board **401** is located, the

first sub-section **4011**A may include a first region **4011**X and a second region **4011**Y with different thicknesses; or, the second sub-section **4011**B may include a first region **4011**X and a second region **4011**Y with different thicknesses; or the first sub-section **4011**A and the second sub-section **4011**B each may include a first region **4011**X and a second region **4011**Y with different thicknesses. The first region **4011**X may be closer to the groove bottom **4010**C than the second region **4011**Y. Along the direction X from the first sub-section **4011**A to the second sub-section **4011**B, the thickness D**1** of the first sub-section **4011**A in the first region **4011**X may be smaller than the thickness D**2** of the first sub-section **4011**A in the second region **4011**Y; and/or, along the direction X from the first sub-section **4011**A to the second sub-section **4011**B, a thickness D**3** of the second sub-section **4011**B in the first region **4011**X may be smaller than a thickness D**4** of the second sub-section **4011**B in the second region **4011**Y. Therefore, the first pin **401** formed by the two independent first sub-section **4011**A and the second sub-section **4011**B may form a relatively large space which is pluggable to a corresponding one of the plurality of second pins **4021** at a position close to the groove bottom **4010**, while a space which is pluggable to the corresponding one of the plurality of second pins **4021** formed near the second connection board **402** may be smaller. The shape of the corresponding one of the plurality of second pins **4021** may match the shape of the first pin **4011**. Correspondingly, after the corresponding second pin **4021** is plugged and connected into the position of the first pin **4011** of the shape, the corresponding second pin **4021** on the second connection board **402** may be firmly embedded into the first pin **4011** of the shape, which may be beneficial to improve the stability of the first connection board **401** and the second connection board **402** after they are plugged and connected.

(51) Also, the space near the second connection board **402** which is pluggable to the corresponding one of the plurality of second pins **4021** formed by the first pin **401** composed of the two independent first sub-section **4011**A and the second sub-section **4011**B may be smaller. Therefore, When the corresponding second pin **4021** is plugged and connected into the first pin **4011**, it may first contact the thicker second region **4011**Y of the first sub-section **4011**A and the second sub-section **4011**B. The plugging and connecting operation may stop in time when misalignment is found, to prevent the corresponding second pin **4021** of the second connection board **402** from contacting the first pin **4011** too much in the groove **4010** and being deeply embedded. The problem that it is difficult to unplug may be avoided, reducing the probability of damage to the pins of the connector **40** during plug and unplug.

(52) In some embodiments, as shown in FIG. **1** to FIG. **7**, the first insulating layer **50** may be located on a surface of the first sub-section **4011**A in the second region **4011**Y; and/or the first insulating layer **50** may be located on a surface of the second sub-section **4011**B in the second region **4011**Y.

(53) In the present embodiment, the first insulating layer **50** coated on the first pin **4011** may be located only on the surface of the first sub-section **4011**A of the second region **4011**Y, or the first insulating layer **50** may be located only on the surface of the second sub-section **4011**B of the second region **4011**Y, or the first insulating layer **50** may be located on the surfaces of the first sub-section **4011**A and the second sub-section **4011**B of the second region **4011**Y. The lower end of a charged second pin **4021** of the plurality of second pins **4021** may first contact the surface of the first sub-section **4011**A and/or the second sub-section **4011**B of the corresponding first pin **4011** in the range of the second region **4011**Y when the second connection board **402** is plugged downward into the first connection board **401**. Therefore, in this embodiment, the first insulating layer **50** may be only applied to the surface of the first sub-section **4011**A of the second region **4011**Y and/or the surface of the second sub-section **4011**B of the second region **4011**Y. Correspondingly, it may be ensured that the final contact range between the first pin **4011** and the corresponding second pin **4021** is large enough to ensure the stability of the electrical connection between the first connection board **401** and the second connection board **402**. Further, even when misplaced and plugged, the first insulating layer **50** on the surface of the first sub-section **4011**A and/or the surface of the

second sub-section **4011**B of the second region **4011**Y may play the role of electrostatic protection to avoid the electrostatic breakdown induced by instantaneous connection and conduction between the first pin **4011** and the corresponding second pin **4021**, thereby preventing electrostatic damage to the display panel **10** during the plug and unplug of the connector **40** with electricity.

(54) In some other embodiments as shown in FIG. **1** to FIG. **5**, FIG. **6** which is a top view of the first connection board in FIG. **4**, and FIG. **8** which is another cross-sectional view along the A-A′ direction in FIG. **6**, along the direction Z**1** from the second region **4011**Y to the first region **4011**X, the thickness of the first sub-section **4011**A in the direction X from the first sub-section **4011**A to the second sub-section **4011**B may increase first, then remain unchanged, and then decrease.

(55) Along the direction Z**1** from the second region **4011**Y to the first region **4011**X, the thickness of the second sub-section **4011**B in the direction X from the first sub-section **4011**A to the second sub-section **4011**B may increase first, then remain unchanged, and then decrease.

(56) In the direction X from the first sub-section **4011**A to the second sub-section **4011**B, a region where the thickness of the first sub-section **4011**A and the second sub-section **4011**B increases may be a first sub-section **4011**Y**1**, and a region where the thicknesses of the first sub-section **4011**A and the second sub-section **4011**B remain unchanged may be a second sub-region **4011**Y**2**.

(57) The first insulating layer **50** may be coated on the first sub-section **4011**A of the first sub-region **4011**Y**1**, and the first insulating layer **50** may be coated on the second sub-section **4011**B of the first sub-region **4011**Y**1**.

(58) In the present embodiment, the first insulating layer **50** coated on the first pin **4011** may be only located on the first sub-section **4011**A of a portion of the second region **4011**Y the first insulating layer **50** coated on the first pin **4011** may be only located on the first sub-section **4011**A of a portion of the second region **4011**Y, or the first insulating layer **50** coated on the first pin **4011** may be only located on the first sub-section **4011**A and the second sub-section **4011**B of a portion of the second region **4011**Y. Along the direction Z**1** from the second region **4011**Y to the first region **4011**X, that is, along the vertical direction from top to bottom in FIG. **8**, the thickness of the first sub-section **4011**A in the direction X from the first sub-section **4011**A to the second sub-section **4011**B may increase first, then remain unchanged, and then decrease. Along the direction Z**1** from the second region **4011**Y to the first region **4011**X, that is, along the vertical direction from top to bottom in FIG. **8**, the thickness of the second sub-section **4011**B in the direction X from the first sub-section **4011**A to the second sub-section **4011**B may increase first, then remain unchanged, and then decrease. The two independent sub-sections of the first pin **4011** each may include a slope area and a vertical area. In this embodiment, the first insulating layer **50** may be coated only on the first sub-section **4011**A of the first sub-region **4011**Y**1** of the slope area, and may be only coated on the second sub-section **4011**B of the first sub-region **4011**Y**1** in the slope area. The lower end of a charged second pin **4021** of the plurality of second pins **4021** may first contact the surface of the slope area of the first sub-section **4011**A and/or the second sub-section **4011**B of the corresponding first pin **4011**. Therefore, in the present embodiment, the first insulating layer **50** may be coated on the first sub-section **4011**A of the first sub-region **4011**Y**1**, and the first insulating layer **50** may be coated on the second sub-section **4011**B of the first sub-region **4011**Y**1**. Correspondingly, it may be ensured that the final contact range between the first pin **4011** and the corresponding second pin **4021** is large enough to ensure the stability of the electrical connection between the first connection board **401** and the second connection board **402**. Further, even when misplaced and plugged, the first insulating layer **50** on the surface of the first sub-section **4011**A and/or the surface of the second sub-section **4011**B of the first sub-region **4011**Y**1** may play the role of electrostatic protection to avoid the electrostatic breakdown induced by instantaneous connection and conduction between the first pin **4011** and the corresponding second pin **4021**, thereby preventing electrostatic damage to the display panel **10** during the plug and unplug of the connector **40** with electricity. Further, the first insulation layer **50** may be only coated on the first pin **4011** in the first sub-region **4011**Y**1**, to save the coating material of the first insulating layer **50** and reduce cost.

(59) For description purposes only, the previous embodiment with the first sub-section **4011**A and the second sub-section **4011**B of the first pin **4011** is used as an example to illustrate the present disclosure, and does not limit the scope of the present disclosure. In various embodiments, the structure of the first pin **4011** may be any suitable structures as long as the corresponding second pin **4021** first contacts the portion of the first pin **4011** coated with the first insulating layer **50** when the first connection board **401** and the second connection board **402** are plugged and connected for electrostatic protection.

(60) As shown in FIGS. **1-5**, FIG. **6** and FIG. **7**, in one embodiment, the first insulating layer **50** may be coated on the first sub-section **4011**A of the second sub-region **4011**Y**2**, and/or, the first insulating layer **50** may be coated on the second sub-section **4011**B of the second sub-region **4011**Y**2**.

(61) Specifically, the first insulating layer **50** coated on the first pin **4011** may only be located on the surface of the first sub-section **4011**A of a partial area of the second region **4011**Y, or the first insulating layer **50** may only be located on the surface of the second sub-section **4011**B of a partial area of the second region **4011**Y, or the first insulating layer **50** may be located on the surfaces of the first sub-section **4011**A and the second sub-section **4011**B of only a partial area of the second region **4011**Y. Along the direction Z**1** from the second region **4011**Y to the first region **4011**X, that is, along the vertical direction from top to bottom in FIG. **8**, the thickness of the first sub-section **4011**A in the direction X from the first sub-section **4011**A to the second sub-section **4011**B may increase first, then remain unchanged, and then decrease. Along the direction Z**1** from the second region **4011**Y to the first region **4011**X, that is, along the vertical direction from top to bottom in FIG. **8**, the thickness of the second sub-section **4011**B in the direction X from the first sub-section **4011**A to the second sub-section **4011**B may increase first, then remain unchanged, and then decrease. The two independent sub-sections of the first pin **4011** each may include a slope area and a vertical area. In this embodiment, the first insulating layer **50** may be coated on the first sub-section **4011**A of the first sub-region **4011**Y**1** in the slope area and the second sub-region **4011**Y**2** in the vertical area, and may be coated the second sub-section **4011**B of the first sub-region **4011**Y**1** in the slope area and the second sub-region **4011**Y**2** in the vertical area. The lower end of a charged second pin **4021** of the plurality of second pins **4021** may first contact the surfaces of the slope area and the vertical area of the first sub-section **4011**A and/or the second sub-section **4011**B of the corresponding first pin **4011**, when the second connection board **402** is plugged into the first connection board **401** downward. Therefore, in the present embodiment, the first insulating layer **50** may be coated on the first sub-section **4011**A of the first sub-region **4011**Y**1** and the second sub-region **4011**Y**2**, and/or on the second sub-section **4011**B of the first sub-region **4011**Y**1** and the second sub-region **4011**Y**2**. Therefore, when misplaced and plugged, the first insulating layer **50** on the surface of the first sub-section **4011**A and/or the surface of the second sub-section **4011**B of the first sub-region **4011**Y**1** and the second sub-region **4011**Y**2** may play the role of electrostatic protection to avoid the electrostatic breakdown induced by instantaneous connection and conduction between the first pin **4011** and the corresponding second pin **4021**, thereby preventing electrostatic damage to the display panel **10** during the plug and unplug of the connector **40** with electricity.

(62) In some optional embodiments, as shown in FIG. **1** to FIG. **8**, the first insulating layer **50** may be made of a material including a metal chelate compound.

(63) Specifically, the first insulating layer **50** may be made of a material including a metal chelate compound. Optionally, in one embodiment, in terms of cost, ease of acquisition, etc., the first insulating layer **50** may be made of an aluminum chelate compound. Metal chelate compounds may be easily obtained by reacting metal alkoxides with chelating agents. In the present disclosure, the first insulating layer **50** may be made of a metal chelate compound, which may improve the adhesion with the metal conductive material of the first pin **4011** in a small area and avoid the falling off of the first insulating layer **50** on the first pin **4011** when the coating area is small. The

coating firmness between the first insulating layer **50** and the first pin **4011** may be ensured.
(64) Optionally, in this embodiment, the thickness of the first insulating layer **50** may be about 0.04 mm to 0.1 mm. When the first insulating layer **50** is made of a metal chelate compound, the coating thickness of the first insulating layer **50** may be ensured to be thin only between about 0.04 mm-0.1 mm, and the coating firmness between the first insulating layer **50** and the first pin **4011** may be improved. Further, the first insulating layer **50** may be prevented from occupying the plugging space between the first pin **4011** and the corresponding second pin **4021**.
(65) In some other embodiments, as shown in FIG. **1**, FIG. **2**, FIG. **9** which is a schematic exploded view of the cross-sectional structure of a first sub-pin and a second sub-pin of the connector in FIG. **2**, and FIG. **10** which is a schematic exploded structure of the cross-sectional structure of a third sub-pin and a fourth sub-pin of the connector in FIG. **2**, the plurality of first pins **4011** may at least include a first sub-pin **40111** and a third sub-pin **40112**. The type of signal connected to the first sub-pin **40111** may be different from the type of the signal connected to the third sub-pin **40112**.
(66) The plurality of second pins **4021** may at least include a second sub-pin **40211** and a fourth sub-pin **40212**. The type of signal connected to the second sub-pin **4021** may be different from the type of signal connected to the fourth sub-pin **40212**.
(67) The second sub-pin **40211** may be connected to the first sub-pin **40111** correspondingly, and the fourth sub-pin **40212** may be connected to the third sub-pin **40112** correspondingly.
(68) As shown in FIG. **9** and FIG. **10**, the shape of the first sub-pin **40111** may be different from the shape of the third sub-pin **40112**.
(69) In the present embodiment, in the first connection board **401** of the connector **40**, the plurality of first pins **4011** may be designed with different shapes. Specifically, the plurality of first pins **4011** may at least include the first sub-pin **40111** and the third sub-pin **40112**, and the signals connected to the first sub-pin **40111** may be different from the type of the signal connected to the third sub-pin **40112**. That is, the plurality of first pins **4011** may include at least two types of different signal pins. The plurality of second pins **4021** in the second connection board **402** may include at least the second sub-pin **40211** corresponding and connected to the first sub-pin **40111** and the fourth sub-pin **40212** corresponding and connected to the third sub-pin **40112**. The plurality of first pins **4011** may be designed with different shapes, and the plurality of second pins **4021** may also be designed with different shapes. Therefore, when the first connection board **401** and the second connection board **402** are plugged and there is a misplaced insertion, the second sub-pin **40211** may not correspond to the position of the first sub-pin **40111**, the fourth sub-pin **40212** may correspond to the position of the first sub-pin **40111**; the fourth sub-pin **40212** may not correspond to the position of the third sub-pin **40112** and the second sub-pin **40211** may correspond to the position of the third sub-pin **40112**. Correspondingly, the first sub-pin **40111** on the first connection board **401** and the fourth sub-pin **40212** on the second connection board **402** which do not match in shape may not be able to form a plug-in structure. That is, the fourth sub-pins **40212** may not be able to be plugged into the positions of the first sub-pins **40111** embedded in the grooves **4010**, thereby avoiding the misaligned plugging of the first connection board **401** and the second connection board **402**.
(70) The present disclosure does not specifically limit the shape of the first sub-pin **40111** and the shape of the third sub-pin **40112** on the first connection board **401**, as long as the shapes of the first sub-pin **40111** on the first connection board **401** and the third sub-pins **40112** on the second connection board **402** are different and cannot match when the first connection board **401** and the second connection board **402** are plugged downward.
(71) Optionally, in one embodiment, the display panel **10** may include light-emitting elements (not shown in the figure), and the light-emitting elements may be understood as sub-pixels, organic light-emitting elements, or micro-light-emitting elements, which is not limited in this embodiment. The light-emitting elements may be connected to a first power supply signal for providing the light-emitting element with a power supply signal for driving light-emitting. For example, the first

power supply signal may be a high-voltage power supply signal. The flexible circuit board **20** and the printed circuit board **30** may be connected to a second power supply signal for providing the flexible circuit board **20** and the printed circuit board **30** with a power supply signal for driving operation. The display panel **10** may also be a touch display panel including a touch structure, that is, the display panel **10** may include touch electrodes (not shown in the figure), and the touch electrodes may be connected to touch drive signals for providing touch electrodes with touch drive signal for the touch-control function. The signals of the first sub-pin **40111** on the first connection board **401** and/or the second sub-pin **40211** on the second connection board **402** with the shape-differentiated design in this embodiment may include any one of the first power signal, the second power signal and the touch driving signal. Since the first power signal, the second power signal, and the touch driving signal are all high-voltage driving signals, in this embodiment, the signal connected to the sub-pins of the connector **40** with shape-differentiated design may be configured to be the high-voltage driving signal. Therefore, the sub-pins connected to the high-voltage signal may be prevented from being damaged by high voltage when the connection board is misaligned and plugged, thereby further reducing the damage to the connector **40**.

(72) In some embodiments, as shown in FIG. **1**, FIG. **2**, FIG. **9** and FIG. **10**, a first protrusion **40111**A may be disposed at a position on the groove bottom **4010**C corresponding to the first sub-pin **40111**.

(73) A side surface of the second sub-pin **40211** facing the first sub-pin **40111** may be provided with a first groove **40211**A.

(74) The first protrusion **40111**A may be plugged into and connected to the first groove **40211**A, and the first protrusion **40111**A and the first groove **40211**A may be matched with each other.

(75) Optionally, in some embodiments, the shapes of the third sub-pin **40112** and the fourth sub-pin **40212** in this embodiment may not include matching protrusion and groove structures, and may be the structure of the pins in the existing technologies. That is, the groove bottom **4010**C at the corresponding position of the three sub-pins **40112** may be a flat surface, and the surface of the fourth sub-pin **40212** facing the first sub-pin **40111** may be also a flat surface. When the first connection board **401** and the second connection board **401** are blindly buckled and there is a misaligned plugging, the fourth sub-pin **40212** on the flat surface may be supported by the first protrusion **40111**A included in the groove bottom **4010**C at the position corresponding to the first sub-pin **40111**, as shown in FIG. **11** which is a schematic diagram of the structure when the first connection board and the second connection board in FIG. **9** and FIG. **10** are dislocated and plugged, and FIG. **12** which is another schematic diagram of the structure when the first connection board and the second connection board in FIG. **9** and FIG. **10** are dislocated and plugged. When the first connection board **401** and the second connection board **402** are misaligned and plugged, the first protrusion **40111**A on the first sub-pin **40111** of the first connection board **401** that is easily damaged with a connected high-voltage signal may support the fourth sub-pin **40212** without the first groove **40211**A (as shown in FIG. **11**), and the second sub-pin **40211** on the second connection board **402** that is easily damaged with a connected high-voltage signal and the third sub-pin **40112** on the first connection board **401** may be suspended in the air (as shown in FIG. **12**) because of the existence of the first groove **40211**A and the support of the first protrusion **40111**A. That is, the vulnerable second sub-pin **40211** connected to the high-voltage signal on the misplaced second connection board **402** may not contact the third sub-pin **40112** on the first connection board **401**, and thus may not be electrically conductive. It may avoid the problem of electrostatic damage caused by the conduction of the vulnerable sub-pins that are connected to the high-potential signal instantly when being wrong plugged, and improve the electrostatic protection effect of the display module **000**.

(76) In the present disclosure, the shapes of the first protrusion **40111**A and the first groove **40211**A are not specifically limited. In one embodiment, the shape of the first protrusion **40111**A may be a trapezoid as shown in FIG. **9**. In some other optional embodiments, the shape of the first protrusion

40111A may be any one or more of a cone, an ellipse, a circle, a triangle, a square, or a rectangle, as long as the groove bottom **4010**C at the position corresponding to the first sub-pin **40111** includes the first protrusion **40111**A. The first sub-pin **40111** and the third sub-pin **40112** may be designed differently, the shape of the first protrusion **40111**A may match the shape of the first groove **40211**A, and when the first sub-pin **40111** and the second sub-pin are plugged in, the first protrusions **40111**A may be plugged into the first grooves **40211**A.

(77) In some other embodiments, as shown in FIG. **1**, FIG. **2**, FIG. **9**, and FIG. **13** which is another schematic exploded view of the cross-sectional structure of a first sub-pin and a second sub-pin of the connector in FIG. **2**, the plurality of first pins may include a plurality of first sub-pins **4-111**, and the plurality of first sub-pins **40111** may at least include a first-type first sub-pin **401111** and a second-type first sub-pin **401112**. The signal connected to the first-type first sub-pin **401111** may be different from the signal connected to the second-type first sub-pin **401112**.

(78) The shape of the first-type first sub-pin **401111** may be different from the shape of the second-type first sub-pin **401112**. Optionally, the shape of the first protrusions **40111**A corresponding to the first-type first sub-pins **401111** may be different from the shape of the first protrusions **40111**A corresponding to the second-type first sub-pins **401112**.

(79) Specifically, in the present embodiment, the plurality of first sub-pins **40111** that are easily damaged by static electricity connected to high-voltage signals may also be designed with different shapes. For example, the plurality of first sub-pins **40111** may at least include the first-type first sub-pin **401111** and the second-type first sub-pin **401112**. The signal connected to the first-type first sub-pin **401111** may be different from the signal connected to the second-type first sub-pin **401112**. Optional, the signal connected to the first-type first sub-pin **401111** may be one of the first power signal, the second power signal, or the touch driving signal among the high-voltage signals, and the signal connected to the second-type first sub-pin **401112** may be other one of the first power signal, the second power signal, or the touch driving signal among the high voltage signals. The shape of the first-type first sub-pin **401111** may be different from the shape of the second-type first sub-pin **401112**. As shown in FIG. **9** and FIG. **13**, the shape of the first protrusions **40111**A corresponding to the first-type first sub-pins **401111** may be a trapezoid, and the shape of the first protrusions **40111**A corresponding to the second-type first sub-pins **401112** may be a semicircle. In some other embodiments as shown in FIG. **1**, FIG. **2**, FIG. **9**, and FIG. **14** which is another schematic exploded view of the cross-sectional structure of a first sub-pin and a second sub-pin of the connector in FIG. **2**, the shape of the first-type first sub-pin **401111** may be different from the shape of the second-type first sub-pin **401112**. Along the direction Z perpendicular to the first connection board **401**, a height H**1** of the first protrusion **40111**A corresponding to the first-type first sub-pin **401111** may be different form a height H**2** of the first protrusion **40111**A corresponding to the second-type first sub-pin **401112**. In some other embodiments as shown in FIG. **1**, FIG. **2**, FIG. **9**, and FIG. **15** which is another schematic exploded view of the cross-sectional structure of a first sub-pin and a second sub-pin of the connector in FIG. **2**, the shape of the first-type first sub-pin **401111** may be different from the shape of the second-type first sub-pin **401112**. Along the direction X parallel to the plane of the first connection board **401**, the width W**1** of the first protrusion **40111**A corresponding to the first-type first sub-pin **401111** may be different from the width W**1** of the first protrusions **40111**A corresponding to the second-type first sub-pins **401112**. Therefore, the shapes of the first protrusions **40111**A of the first sub-pins **40111** connected to different high-voltage signals may be designed differently, to differentiate the first sub-pins **40111** connected to different high-voltage signals. The connection accuracy between the first connection board **401** and the second connection board **402** may be improved.

(80) In some embodiments, as shown in FIG. **1**, FIG. **2**, FIG. **9**, and FIG. **16** which is another schematic exploded view of the cross-sectional structure of a third sub-pin and a fourth sub-pin of the connector in FIG. **2**, a second groove **40112**B may be disposed at a position on the groove bottom **4010**C corresponding to the third sub-pin **40112**, and, a first protrusion **40111**A may be

disposed at a side surface of the fourth sub-pin **40212** facing the third sub-pin **40112**.
(81) The third protrusion **40212**B may be plugged into and connected to the second groove **40112**B, and third protrusion **40212**B and the second groove **40112**B may be matched with each other.
(82) Specifically, the signals connected to the third sub-pin **40112** of the first connection board **401** and the fourth sub-pin **40212** of the second connection board **402** correspondingly plugged with the first connection board **401** may be low-voltages, and the third sub-pin **40112** of the first connection board **401** and the fourth sub-pin **40212** of the second connection board **402** may be sub-pins that are not easily damaged by static electricity. The shape of the third sub-pin **40112** and the fourth sub-pin **40212** may also include matching groove and protrusion. That is, the second groove **40112**B may be disposed at the position on the groove bottom **4010**C corresponding to the third sub-pin **40112**, and the first protrusion **40111**A may be disposed at the side surface of the fourth sub-pin **40212** facing the third sub-pin **40112**. When the first connection board **401** and the second connection board **401** are correctly plugged, the first protrusion **40111**A of the first sub-pin **40111** may be plugged into the first groove **40211**A of the second sub-pin **40211**, and the second groove **40112**B of the third sub-pin **40112** may be plugged into the second protrusion **40212**B of the fourth sub-pin **40212**. When the first connection board **401** and the second connection board **401** are blindly buckled and there is a misaligned plugging, the fourth sub-pin **40212** with the second protrusion **40212**B may be supported by the first protrusion **40111**A included in the groove bottom **4010**C at the position corresponding to the first sub-pin **40111**, as shown in FIG. **17** which is a schematic diagram of the structure when the first connection board and the second connection board in FIG. **9** and FIG. **16** are dislocated and plugged, and FIG. **18** which is another schematic diagram of the structure when the first connection board and the second connection board in FIG. **9** and FIG. **16** are dislocated and plugged. When the first connection board **401** and the second connection board **402** are misaligned and plugged, the first protrusion **40111**A on the first sub-pin **40111** of the first connection board **401** that is easily damaged with a connected high-voltage signal may support the fourth sub-pin **40212** with the second protrusion **40212**B (as shown in FIG. **17**), and the second sub-pin **40211** on the second connection board **402** that is easily damaged with a connected high-voltage signal and the third sub-pin **40112** on the first connection board **401** may be suspended in the air (as shown in FIG. **18**) because of the existence of the first groove **40211**A and the second groove **40112**B, and also the support of the first protrusion **40111**A and the second protrusion **40212**B. That is, the vulnerable second sub-pin **40211** connected to the high-voltage signal on the misplaced second connection board **402** may not contact the third sub-pin **40112** on the first connection board **401**, and thus may not be electrically conductive. It may avoid the problem of electrostatic damage caused by the conduction of the vulnerable sub-pins that are connected to the high-potential signal instantly when being wrong plugged, and improve the electrostatic protection effect of the display module **000**.
(83) In some embodiments, as shown in FIG. **1**, FIG. **2**, FIG. **19** which is another schematic exploded view of the cross-sectional structure of a first sub-pin and a second sub-pin of the connector in FIG. **2**, and FIG. **20** which is another schematic exploded view of the cross-sectional structure of a third sub-pin and a fourth sub-pin of the connector in FIG. **2**, among the plurality of second pins **4021**, the volumes of the second sub-pin **40211** and the fourth sub-pin **40212** may be different.
(84) In the present embodiment, among the plurality of second pins **4021** of the second connection board **402**, the volumes of the second sub-pin **40211** and the fourth sub-pin **40212** may be set to be different, to distinguish between different types of second pins **4021**. For example, the second sub-pin **40211** connected to a high-potential signal and vulnerable to static electricity, and the fourth sub-pin **40212** connected to a low-potential signal and difficult to be damaged by static electricity, may have different volumes. Optionally, the volume of the second sub-pin **40211** may be set to be larger than the volume of the fourth sub-pin **40212**. Therefore, the space for plugging of the first

sub-pin **40111** on the first connection board **401** that matches the second sub-pin **40211** may be larger, and the space for plugging of the third sub-pin **40112** on the first connection board **401** that matches the fourth sub-pin **40212** may be small, such that the fool-proof design for accessing different types of second pins **4021** may be achieved. When the second sub-pin **40211** with a larger volume is misaligned and plugged into the position of the third sub-pin **40112** with a smaller pluggable space, the second sub-pin **40211** that is easily damaged by static electricity and is connected to a high-potential signal may be unable to be plugged into the position of third sub-pin **40112** with a smaller pluggable space. Therefore, a conductive path may not be formed between the second sub-pin **40211** that is easily damaged by static electricity and is connected to the high-potential signal and the first connection board **401**, thereby protecting the second sub-pin **40211** which is easily damaged by static electricity and avoiding the problem of electrostatic damage caused by the conduction of the vulnerable sub-pins that are connected to the high-potential signal instantly when being wrong plugged. The connector **40** may be further protected from the electrostatic damage and the electrostatic protection effect of the display module **000** may be improved.

(85) In the present embodiment, as shown in FIG. **19** and FIG. **20**, that the volumes of the second sub-pin **40211** and the fourth sub-pin **40212** are different may be specifically reflected as that the overall size of the fourth sub-pin **40212** in FIG. **20** may be smaller than that of the second sub-pin **40211** in FIG. **19**. In some other optional embodiments, as shown in FIG. **19** and FIG. **21** which is another exploded structure of the cross-sectional structure of the third sub-pin and the fourth sub-pin of the connector in FIG. **2**, in the direction X parallel to the plane where the first connection board **401** is located, the thickness D**5** of the second sub-pin **40211** may be larger than the thickness D**6** of the fourth sub-pin **40212**. Therefore, in the direction X parallel to the plane of the first connection board **401**, the pluggable width of the first sub-pin **40111** on the first connection board **401** that matches the second sub-pin **40211** may be larger, and the pluggable width of the third sub-pin **40112** on the first connection board **401** that matches the fourth sub-pin **40212** may be smaller, such that the first sub-pin **40111** on the first connection board **401** and the second sub-pin **40211** on the second connection board **402** may be able to be matched and plugged, and the third sub-pin **40112** on the first connection board **401** and the fourth sub-pin **40212** on the second connection board **402** may be able to be matched and plugged.

(86) In some other optional embodiments, as shown in FIG. **1**, FIG. **2**, FIG. **19** and FIG. **22** which is another exploded view of the cross-sectional structure of the third sub-pin and the fourth sub-pin of the connector in FIG. **2**, in the direction Z perpendicular to the plane where the first connection board **401** is located, the height H**3** of the second sub-pin **40211** may be smaller than the height H**4** of the fourth sub-pin **40212**. Therefore, in the direction Z perpendicular to the plane where the first connection board **401** is located, the depth of the groove **4010** at the position corresponding to the first sub-pin **40111** on the first connection board **401** that matches the second sub-pin **40211** may be smaller than the depth of the groove **4010** at the position corresponding to the third sub-pin **40112** on the first connection board **401** that matches the fourth sub-pin **40212**. In the present embodiment, that the volumes of the second sub-pin **40211** and the fourth sub-pin **40212** are different may also be reflected that the height H**3** of the second sub-pin **40211** may be smaller than the height H**4** of the fourth sub-pin **40212** in the direction Z perpendicular to the plane where the first connection board **401** is located. That is, the height of the second sub-pin **40211** that is easily damaged by static electricity and connected to the high-potential signal may be lower, and the depth of the groove **4010** at the position corresponding to the matching first sub-pin **40111** on the first connection board **401** may be also shallower. When the second sub-pin **40211** with the lower height is plugged into the third sub-pin **40112** on the first connection board **401** with the deeper depth in misaligned plugging, since the height H**3** of the second sub-pin **40211** is smaller than the height H**4** of the fourth sub-pin **40212**, the depth of the groove **4010** at the corresponding position of the third sub-pin **40112** may be deeper, and the second sub-pin **40211** may not contact the

conductive part of the third sub-pin **40112** to conduct. Therefore, a conductive path may not be formed between the second sub-pin **40211** that is easily damaged by static electricity and is connected to the high-potential signal and the first connection board **401**, thereby protecting the second sub-pin **40211** which is easily damaged by static electricity and avoiding the problem of electrostatic damage caused by the conduction of the vulnerable sub-pins that are connected to the high-potential signal instantly when being wrong plugged. The connector **40** may be further protected from the electrostatic damage and the electrostatic protection effect of the display module **000** may be improved.
(87) In some optional embodiments, as shown in FIG. **23** which is another schematic plan view of the display module and FIG. **24** which is a connection structure of the flexible circuit board and the printed circuit board in FIG. **23**, the bottom of the first connection board **401** away from the second connection board **402** may include a plurality of first conductive pads **601**, and the first connection board **401** may be bonded and electrically connected to the printed circuit board through the plurality of first conductive pads **601**.
(88) The bottom of the second connection board **402** away from the first connection board **401** may include a plurality of second conductive pads **602**, and the second connection board **402** may be bonded and electrically connected to the flexible circuit board **20** through the plurality of second conductive pads **602**.
(89) In the present embodiment, when the flexible circuit board **20** and the printed circuit board **30** are directly plugged to achieve electrical connection through the connector **40**, the bottom of the first connection board **401** away from the second connection board **402** may include the plurality of first conductive pads **601**, and the bottom of the second connection board **402** away from the first connection board **401** may include the plurality of second conductive pads **602**. The first connection board **401** may be bonded and electrically connected to the printed circuit board through the plurality of first conductive pads **601** at the corresponding position, and the second connection board **402** may be bonded and electrically connected to the flexible circuit board **20** through the plurality of second conductive pads **602** at the corresponding position. Therefore, the electrical conduction between the flexible circuit board **20** and the printed circuit board **30** may be achieved by the electrical contacts through pin plugging of the first connection board **401** and the second connection board **402** of the connector **40**, such that the driving signal provided by the printed circuit board **30** is transmitted to the flexible circuit board **20** through the pins of the connector **40**. The flexible circuit board **20** may be electrically bonded connected to the display panel **10** in the bonding area BA of the display panel **10**, such that the driving signal of the printed circuit board **30** may be provided to the display panel **10** for providing the display panel **10** with a driving signal for realizing functions such as display.
(90) In one embodiment, each of the plurality of first conductive pads **601** on the first connection board **401** and a corresponding one of the plurality of first pins **4011** may be electrically connected correspondingly inside the first connection board **401**, and each of the plurality of second conductive pad **602** on the second conduction board **402** and a corresponding one of the plurality of second pins **4021** may be electrically connected correspondingly inside the second connection board **402** (the connecting circuit layer inside the connection board is not shown in the figure). A connection board itself may be understood as a structure including a circuit layer, and the present disclosure does not limit the circuit structure inside the connector **40**, as long as the flexible circuit board **20** and the printed circuit board **30** are able to realize signal transmission through the connector **40**.
(91) In some optional embodiments, as shown in FIG. **23** and FIG. **25** which is a schematic diagram of another connection structure of the flexible circuit board and the printed circuit board in FIG. **23**, the bottom of the first connection board **401** away from the second connection board **402** may include gate switches **70**, and the gate switches **70** may be configured to control the plurality of first pins **4011** of the first connection board **401** to be conductive or non-conductive.

(92) In the present embodiment, the connector **40** for plugging and electrically connecting the flexible circuit board **20** and the printed circuit board **30** may be provided with the gate switches **70**. Optionally, the gate switch **70** may be provided on the bottom of the first connection board **401** away from the second connection board **402**. Two or more gate switches **70** may be respectively disposed on the bottom of two ends of the first connection board **401**, and the gate switches **70** may be used to control the plurality of first pins **4011** of the first connection boards **401** to be conductive or non-conductive. For example, when the gate switches **70** are pressed, the plurality of first pins **4011** on the first connection board **401** and the internal circuit layer of the first connection board **401** may be electrically connected, and the first pins **4011** may carry electricity. When the gate switches **70** are not pressed, the plurality of first pins **4011** on the first connection board **401** and the internal circuit layer of the first connection board **401** may be not conductive, and the plurality of first pins **4011** may not be charged. At this time, when the operator makes mistakes in misaligned plugging, the electrostatic damage caused by the misaligned plugging contact may be avoided, thereby protecting the connector **40** from electrostatic damage and improving the electrostatic protection effect of the display module **000**.
(93) As shown in FIG. **23** and FIG. **26** which is a schematic diagram of the three-dimensional exploded structure of the connector in FIG. **23**, in one embodiment, the gate switches **70** may be disposed at the bottom of the groove **4010** opened on the first connection board **401**. When the plurality of second pins **4021** of the second connection board **402** is correctly plugged into the groove **4010** of the first connection board **401**, a bump **80** of the second connection board **401** may press the gate switches **70** at the corresponding position in the groove **4010**, the plurality of first pins **4011** on the first connection board **401** may be electrically connected to the internal circuit layer of the first connection board **401**, and the plurality of first pins **4011** may be charged. When the plurality of second pins **4021** of the second connection board **402** are misaligned and plugged into the groove **4010** of the first connection board **401**, the bump **80** of the second connection board **401** may not press the gate switches **70** at the corresponding position in the groove **4010**, then the gate switches **70** may not be pressed. Correspondingly, the plurality of first pins **4011** and the internal circuit layer of the first connection board **401** may be not conductive, and the plurality of first pins **4011** may be not charged. That is, when the first connection board **401** and the second connection board **402** are misaligned and plugged, the gate switches **70** may avoid electrostatic damage caused by misaligned and plugged contacts, thereby protecting the connector **40** from electrostatic damage and improving the electrostatic protection effect of the display module **000**.
(94) It can be understood that the present disclosure does not specifically limit the design structure and setting position of the gate switches **70**, and the embodiments shown in FIG. **25** and FIG. **26** are only for illustration. During the specific implementation, the setting position of the gate switches **70** is not limited to this, and may be performed according to actual design requirements, which will not be repeated in this embodiment.
(95) The present disclosure also provides a display device. As shown in FIG. **27**, in one embodiment, the display device **111** may include any display panel **000** provided by various embodiments of the present disclosure. The embodiment shown in FIG. **27** where the display device **111** is a cell phone is used as an example only to illustrate the present disclosure, and does not limit the scopes of the present disclosure. In some other embodiments, the display device **111** may be any display device with display function, such as a computer, a television, a touch controller, or a vehicle display device.
(96) In the present disclosure, the display module may include the display panel, the flexible circuit board and the printed circuit board that are bound and electrically connected to the display panel. The flexible circuit board and the printed circuit board may be bonded and electrically connected to the display panel in the bonding area of the display panel. The flexible circuit board and the printed circuit board may be connected through the connector, such that the printed circuit board may provide the display panel with electrical signals that drive it to realize functions such as display

through the circuits on the flexible circuit board. The connector may include the first connection board and the second connection board. The side of the first connection board facing the second connection board may include the groove, and the groove may include the plurality of first pins. The side of the second connection board facing the first connection board may include the plurality of second pins. The plurality of first pins and the plurality of second pins may have a one-to-one correspondence. The electrical connection between the display panel and the printed circuit board may be easily and quickly completed by plugging and unplugging between the first connection board and the second connection board of the connector. The industrialized mass production of the module may be achieved. Further, a partial area of at least one of the plurality of first pins in the first connection board may be coated with the first insulating layer. When the flexible circuit board and the printed circuit board are electrically connected through the connector, the plurality of second pins of the second connection board may be plugged into the groove of the first connection board, and electrically connected to the corresponding first pins. Since there is the first insulating layer coated in part of the plurality of first pins, the electrostatic breakdown because of the instantaneous contact between the plurality of first pins and the plurality of second pins during the instantaneous misaligned plugging, thereby preventing electrostatic damage to the display panel during the plugging operation with electricity. Also, since the first insulating layer is not coated on the entire surface of the plurality of first pins, the first insulating layer may not affect the conductivity of the plurality of first pins, ensuring that the plurality of second pins and the part of the plurality of first pins not covered by the first insulating layer are still able to be in contact to achieve the conductive transmission between the first connection board and the second connection board when the first connection board and the second connection board are plugged.
(97) Various embodiments have been described to illustrate the operation principles and exemplary implementations. It should be understood by those skilled in the art that the present disclosure is not limited to the specific embodiments described herein and that various other obvious changes, rearrangements, and substitutions will occur to those skilled in the art without departing from the scope of the disclosure. Thus, while the present disclosure has been described in detail with reference to the above described embodiments, the present disclosure is not limited to the above described embodiments, but may be embodied in other equivalent forms without departing from the scope of the present disclosure, which is determined by the appended claims.

Claims

1. A display module, comprising a display panel, a flexible circuit board, and a printed circuit board, wherein: the display panel includes a bonding area; the flexible circuit board and the printed circuit board are electrically bonded and connected to the display panel in the bonding area; the flexible circuit board and the printed circuit board are connected by a connector; the connector includes a first connection board and a second connection board; a side of the first connection board facing the second connection board includes a groove; the groove includes a plurality of first pins; a side of the second connection board facing the first connection board includes a plurality of second pins; the plurality of first pins and the plurality of second pins have a one-to-one correspondence; the plurality of second pins is plugged into the grooves, and is in contact with and electrically connected to the plurality of first pins; and a portion of at least one of the plurality of first pins is coated with a first insulating layer.
2. The display module according to claim 1, wherein: a shape of an orthographic projection of the groove on the plane where the first connection board is located includes a long strip; the groove includes a first side wall and a second side wall; one first pin of the plurality of first pins includes a first sub-section and a second sub-section, wherein the first sub-section is arranged on the first side wall and the second sub-section is arranged on the second side wall; in a direction perpendicular to the plane where the first connection board is located, the first sub-section and/or the second sub-

section includes a first region and a second region, wherein the first region is located at a side of the second region close to a bottom of the groove; and in a direction from the first sub-section to the second sub-section, a thickness of the first sub-section of the first region is smaller than a thickness of the first sub-section of the second region; and/or, in the direction from the first sub-section to the second sub-section, a thickness of the second sub-section of the first region is smaller than a thickness of the second sub-section of the second region.

3. The display module according to claim 2, wherein: the first insulating layer is located on a surface of the first sub-section of the second region; and/or the first insulating layer is located on a surface of the second sub-section of the second region.

4. The display module according to claim 2, wherein: in a direction from the second region to the first region, a thickness of the first sub-section in the direction from the first sub-section to the second sub-section increases first, then does not change, and then decreases; in the direction from the second region to the first region, a thickness of the second sub-section in the direction from the first sub-section to the second sub-section increases first, then does not change, and then decreases; in the direction from first sub-section to the second sub-section, a region where the thicknesses of the first sub-section and the second sub-section increase is a first sub-region, and a region where the thicknesses of the first sub-region and the second sub-region remain unchanged is a second sub-region; and the first insulating layer is coated on the first sub-section in the first sub-region and on the second sub-section of the first sub-region.

5. The display module according to claim 4, wherein: the first insulating layer is coated on the first sub-section in the second sub-region and on the second sub-section of the second sub-region.

6. The display module according to claim 1, wherein: the first insulating layer is made of a material including a metal chelate compound, and/or a thickness of the first insulating layer is about 0.04 mm to about 0.1 mm.

7. The display module according to claim 1, wherein: the plurality of the first pins at least includes first sub-pins and third sub-pins, wherein signals connected to the first sub-pins and signals connected to the third sub-pins have different types; the plurality of second pin includes second sub-pins and fourth sub-pins, wherein signals connected to the second sub-pins and signals connected to the fourth sub-pins have different types; the second sub-pins are plugged into and correspond to the first sub-pins, and the fourth sub-pins are plugged into and correspond to the third sub-pins; and shapes of the first sub-pins are different from shapes of the third sub-pins.

8. The display module according to claim 7, wherein: the display panel includes a light-emitting element, wherein the light-emitting element is connected to a first power supply signal; the flexible circuit board and the printed circuit board are connected to a second power supply signal; the display panel includes a touch electrode, wherein the touch electrode is connected to a touch control drive signal; the signals connected to the first sub-pins and/or the second sub-pins include any one of the first power supply signal, the second power supply signal, or the touch control drive signal.

9. The display module according to claim 7, wherein: a first protrusion is disposed on the bottom of the groove at a position corresponding to the first sub-pins; a side surface of the second sub-pins facing the first sub-pin includes a first groove; the first sub-pins are plugged into and connected to the first groove; and the first protrusion and the first groove are matched with each other.

10. The display module according to claim 9, wherein: the first sub-pins include at least a first-type first sub-pin and a second-type first sub-pin, wherein signals connected to the first-type first sub-pin and the second-type first sub-pin are different; and a shape of the first-type first sub-pin is different form a shape of the second-type first sub-pin.

11. The display module according to claim 10, wherein: in the direction perpendicular to the plane where the first connection board is located, a height of the first protrusion corresponding to the first-type first sub-pin is different from a height of the first protrusion corresponding to the second-type first sub-pin; or in a direction parallel to the plane where the first connection board is located,

a width of the first protrusion corresponding to the first-type first sub-pin is different from a width of the first protrusion corresponding to the second-type first sub-pin.

12. The display module according to claim 9, wherein: shapes of the first groove/the first protrusion include one or more of cones, trapezoids, ovals, circles, triangles, squares, or rectangles.

13. The display module according to claim 9, wherein: a second groove is disposed on the bottom of the groove at a position corresponding to the third sub-pins; a side surface of the fourth sub-pins facing the third sub-pin includes a second protrusion; the second protrusion is plugged and connected to the second groove; and the second protrusion and the second groove are matched with each other.

14. The display module according to claim 7, wherein: in the plurality of second pins, a volume of the second sub-pins is different from a volume of the fourth sub-pins.

15. The display module according to claim 14, wherein: the volume of the second sub-pins is larger than the volume of the fourth sub-pins.

16. The display module according to claim 14, wherein: in the direction parallel to the plane where the first connection board is located, a thickness of the second sub-pins is larger than a thickness of the fourth sub-pins.

17. The display module according to claim 14, wherein: in the direction perpendicular to the plane where the first connection board is located, a height of the second sub-pins is smaller than a height of the fourth sub-pins; and in the direction perpendicular to the plane where the first connection board is located, a depth of the groove at the position corresponding to the first sub-pins is smaller than a depth of the groove at the position corresponding to the third sub-pins.

18. The display module according to claim 1, wherein: a bottom of the first connection board away from the second connection board includes a plurality of first conductive pads, wherein the first connection board is electrically bonded and connected to the printed circuit board through the plurality of first conductive pads; and a bottom of the second connection board away from the first connection board includes a plurality of second conductive pads, wherein the second connection board is electrically bonded to the flexible circuit board through the plurality of second conductive pads connect.

19. The display module according to claim 1, wherein: a gating switch is disposed at a bottom of the first connection board away from the second connection board, and is used to control the plurality of first pins of the first connection board to conduct or not conduct electricity.

20. A display device, comprising a display module including a display panel, a flexible circuit board, and a printed circuit board, wherein: the display panel includes a bonding area; the flexible circuit board and the printed circuit board are electrically bonded and connected to the display panel in the bonding area; the flexible circuit board and the printed circuit board are connected by a connector; the connector includes a first connection board and a second connection board; a side of the first connection board facing the second connection board includes a groove; the groove includes a plurality of first pins; a side of the second connection board facing the first connection board includes a plurality of second pins; the plurality of first pins and the plurality of second pins have a one-to-one correspondence; the plurality of second pins is plugged into the grooves, and is in contact with and electrically connected to the plurality of first pins; and a portion of at least one of the plurality of first pin is coated with a first insulating layer.